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If, If-else , if-else-if,Nested if, Switch

Question no 1: Easy `if` Programs (1 to 5)

1. Check Positive Number:

```
#include <iostream>
using namespace std;

int main() {
    int number = 10;
    if (number > 0) {
        cout << "Positive Number";
    }
    return 0;
}
```

2. Check Equal to 10:

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    if (x == 10) {
        cout << "Number is 10";
    }
    return 0;
}
```

3. Check Less Than 50:

```
#include <iostream>
using namespace std;

int main() {
    int val = 25;
    if (val < 50) {
     cout << "Value is smaller than 50";
```

```
    }
    return 0;
}
```

4. Check Non-Zero Value:

```
#include <iostream>
using namespace std;

int main() {
    int num = 7;
    if (num != 0) {
        cout << "Number is not zero";
    }
    return 0;
}
```

5. Check Greater Than 100:

```
#include <iostream>
using namespace std;

int main() {
    int mark = 120;
    if (mark > 100) {
        cout << "High Score";
    }
    return 0;
}
```

B. Normal `if` Programs (6 to 10)

6. Check Voting Age:

```
#include <iostream>
using namespace std;

int main() {
    int age = 20;
    if (age >= 18) {
cout << "You are eligible to vote";
    }
    return 0;
}
```

7. Check Passing Marks:

```
#include <iostream>
using namespace std;

int main() {
   int score = 55;
   if (score >= 50) {
     cout << "You have passed";
   }
   return 0;
}
```

8. Check Even Number:

```
#include <iostream>
using namespace std;

int main() {
   int num = 8;
   if (num % 2 == 0) {
      cout << "Even Number";
   }
   return 0;
}
```

9. Check Multiple of 5:

```
#include <iostream>
using namespace std;

int main() {
   int n = 25;
   if (n % 5 == 0) {
      cout << "Multiple of 5";
   }
   return 0;
}
```

10. Check Temperature High:

```
C++
#include <iostream>
using namespace std;

int main() {
   float temp = 38.5;
```

```
    if (temp > 37.0) {
        cout << "Fever";
    }
    return 0;
}
```

C. Advanced `if` Programs (11 to 15)

11. Check Uppercase Character:

C++
```
#include <iostream>
using namespace std;

int main() {
    char ch = 'G';
    if (ch >= 'A' && ch <= 'Z') {
        cout << "Capital Letter";
    }
    return 0;
}
```

12. Check Divisible by 3 and 7:

C++
```
#include <iostream>
using namespace std;

int main() {
    int a = 21;
    if (a % 3 == 0 && a % 7 == 0) {
        cout << "Divisible by 3 and 7";
    }
    return 0;
}
```

13. Check Leap Year (Basic):

C++
```
#include <iostream>
using namespace std;

int main() {
    int year = 2024;
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
```

```
        cout << "Leap Year";
    }
    return 0;
}
```

14. Check Range (10 to 50):

C++
```
#include <iostream>
using namespace std;

int main() {
    int x = 30;
    if (x >= 10 && x <= 50) {
        cout << "In Range";
    }
    return 0;
}
```

15. Check Triangle Validity:

C++
```
#include <iostream>
using namespace std;

int main() {
    int a = 60, b = 60, c = 60;
    if (a + b + c == 180) {
        cout << "Valid Triangle";
    }
    return 0;
}
```

SECTION 2: `if-else` Programs

A. Easy `if-else` Programs (16 to 20)

16. Positive or Negative:

C++
```
#include <iostream>
using namespace std;

int main() {
```

```
    int n = -5;
    if (n >= 0) {
        cout << "Positive";
    } else {
        cout << "Negative";
    }
    return 0;
}
```

17. Greater of Two Numbers:

C++

```
#include <iostream>
using namespace std;

int main() {
    int a = 15, b = 20;
    if (a > b) {
        cout << "A Big";
    } else {
        cout << "B Big";
    }
    return 0;
}
```

18. Equal or Not Equal:

C++

```
#include <iostream>
using namespace std;

int main() {
    int x = 5, y = 5;
    if (x == y) {
        cout << "Equal";
    } else {
        cout << "Not Equal";
    }
    return 0;
}
```

19. Check Zero or Non-Zero:

C++

```
#include <iostream>
using namespace std;
```

```
int main() {
    int val = 0;
    if (val == 0) {
        cout << "Zero";
    } else {
        cout << "Not Zero";
    }
    return 0;
}
```

20. Check Pass/Fail:

C++

```
#include <iostream>
using namespace std;

int main() {
    int marks = 33;
    if (marks >= 40) {
        cout << "Pass";
    } else {
        cout << "Fail";
    }
    return 0;
}
```

B. Normal `if-else` Programs (21 to 25)

21. Odd or Even:

C++

```
#include <iostream>
using namespace std;

int main() {
    int num = 13;
    if (num % 2 == 0) {
        cout << "Even";
    } else {
        cout << "Odd";
    }
    return 0;
}
```

22. Eligible for Driving License:

C++

```
#include <iostream>
using namespace std;

int main() {
    int age = 16;
    if (age >= 18) {
        cout << "Fit Drive";
    } else {
        cout << "No Fit Drive";
    }
    return 0;
}
```

23. Maximum of Two Quantities:

C++

```
#include <iostream>
using namespace std;

int main() {
    double price1 = 99.5, price2 = 120.0;
    if (price1 > price2) {
        cout << "Price 1 High";
    } else {
        cout << "Price 2 High";
    }
    return 0;
}
```

24. Check Divisibility by 10:

C++

```
#include <iostream>
using namespace std;

int main() {
    int n = 45;
    if (n % 10 == 0) {
        cout << "Divisible by 10";
    } else {
        cout << "Not Divisible by 10";
    }
    return 0;
```

```
}
```

25. Vowel or Consonant (Simple):

```
C++
#include <iostream>
using namespace std;

int main() {
    char ch = 'a';
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
        cout << "Vowel";
    } else {
        cout << "Consonant";
    }
    return 0;
}
```

C. Advanced `if-else` Programs (26 to 30)

26. Leap Year Check:

```
C++
#include <iostream>
using namespace std;

int main() {
    int y = 2023;
    if ((y % 4 == 0 && y % 100 != 0) || (y % 400 == 0)) {
        cout << "Leap Year";
    } else {
        cout << "Not Leap Year";
    }
    return 0;
}
```

27. Alphabet or Digit:

```
C++
#include <iostream>
using namespace std;

int main() {
    char ch = '9';
    if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
```

```
        cout << "Alphabet";
    } else {
        cout << "Not Alphabet";
    }
    return 0;
}
```

28. Discount Eligibility:

```
C++
#include <iostream>
using namespace std;

int main() {
    int amount = 1200;
    if (amount >= 1000) {
        cout << "Discount Granted";
    } else {
        cout << "No Discount";
    }
    return 0;
}
```

29. Check Valid Password Length:

```
C++
#include <iostream>
using namespace std;

int main() {
    int len = 7;
    if (len >= 8) {
        cout << "Strong Password";
    } else {
        cout << "Weak Password";
    }
    return 0;
}
```

30. Quadrant Check (Basic):

```
C++
#include <iostream>
using namespace std;

int main() {
```

```
    int x = 5, y = -2;
    if (x > 0 && y > 0) {
        cout << "First Quadrant";
    } else {
        cout << "Other Quadrant";
    }
    return 0;
}
```

SECTION 3: `if-else-if` Programs

A. Easy `if-else-if` Programs (31 to 35)

31. Positive, Negative, or Zero:

C++
```
#include <iostream>
using namespace std;

int main() {
    int num = 0;
    if (num > 0) {
        cout << "Positive";
    } else if (num < 0) {
        cout << "Negative";
    } else {
        cout << "Zero";
    }
    return 0;
}
```

32. Simple Grading (3 Levels):

C++
```
#include <iostream>
using namespace std;

int main() {
    int marks = 75;
    if (marks >= 80) {
        cout << "Grade A";
    } else if (marks >= 50) {
        cout << "Grade B";
```

```
    } else {
        cout << "Grade C";
    }
    return 0;
}
```

33. Compare Two Numbers (3 Cases):

C++

```
#include <iostream>
using namespace std;

int main() {
    int a = 10, b = 20;
    if (a > b) {
        cout << "A Big";
    } else if (b > a) {
        cout << "B Big";
    } else {
        cout << "Both Equal";
    }
    return 0;
}
```

34. Speed Check:

C++

```
#include <iostream>
using namespace std;

int main() {
    int speed = 80;
    if (speed > 100) {
        cout << "Too Fast";
    } else if (speed >= 40) {
        cout << "Normal Speed";
    } else {
        cout << "Slow";
    }
    return 0;
}
```

35. Number Size Category:

C++

```
#include <iostream>
```

```
using namespace std;

int main() {
    int x = 500;
    if (x < 10) {
        cout << "1 Digit";
    } else if (x < 100) {
        cout << "2 Digits";
    } else {
        cout << "3 or More Digits";
    }
    return 0;
}
```

B. Normal `if-else-if` Programs (36 to 40)

36. Weekday Name (1-3):

C++
```
#include <iostream>
using namespace std;

int main() {
    int day = 2;
    if (day == 1) {
        cout << "Monday";
    } else if (day == 2) {
        cout << "Tuesday";
    } else if (day == 3) {
        cout << "Wednesday";
    } else {
        cout << "Other Day";
    }
    return 0;
}
```

37. Age Group Classification:

C++
```
#include <iostream>
using namespace std;

int main() {
    int age = 25;
    if (age < 13) {
        cout << "Child";
```

```
    } else if (age < 20) {
        cout << "Teenager";
    } else if (age < 60) {
        cout << "Adult";
    } else {
        cout << "Senior";
    }
    return 0;
}
```

38. Temperature Level:

C++

```
#include <iostream>
using namespace std;

int main() {
    int t = 32;
    if (t > 40) {
        cout << "Very Hot";
    } else if (t >= 20) {
        cout << "Warm";
    } else if (t >= 0) {
        cout << "Cold";
    } else {
        cout << "Freezing";
    }
    return 0;
}
```

39. Largest of Three Numbers:

C++

```
#include <iostream>
using namespace std;

int main() {
    int a = 12, b = 25, c = 9;
    if (a >= b && a >= c) {
        cout << "A Largest";
    } else if (b >= a && b >= c) {
        cout << "B Largest";
    } else {
        cout << "C Largest";
    }
    return 0;
```

```
}
```

40. Simple Calculator Operation Selection:

```
C++
#include <iostream>
using namespace std;

int main() {
    char op = '+';
    int x = 10, y = 5;
    if (op == '+') {
        cout << x + y;
    } else if (op == '-') {
        cout << x - y;
    } else if (op == '*') {
        cout << x * y;
    } else {
        cout << "Invalid Operator";
    }
    return 0;
}
```

C. Advanced `if-else-if` Programs (41 to 45)

41. Complete Student Grading System:

```
C++
#include <iostream>
using namespace std;

int main() {
    int score = 87;
    if (score >= 90) {
        cout << "Grade A+";
    } else if (score >= 80) {
        cout << "Grade A";
    } else if (score >= 70) {
        cout << "Grade B";
    } else if (score >= 60) {
        cout << "Grade C";
    } else if (score >= 50) {
        cout << "Grade D";
    } else {
        cout << "Grade F";
    }
```

```
    return 0;
}
```

42. Electricity Bill Calculation:

C++

```
#include <iostream>
using namespace std;

int main() {
    int units = 250;
    if (units <= 100) {
        cout << "Bill: " << units * 5;
    } else if (units <= 200) {
        cout << "Bill: " << units * 8;
    } else if (units <= 300) {
        cout << "Bill: " << units * 10;
    } else {
        cout << "Bill: " << units * 12;
    }
    return 0;
}
```

43. Character Type Classifier:

C++

```
#include <iostream>
using namespace std;

int main() {
    char ch = '#';
    if (ch >= 'A' && ch <= 'Z') {
        cout << "Uppercase Letter";
    } else if (ch >= 'a' && ch <= 'z') {
        cout << "Lowercase Letter";
    } else if (ch >= '0' && ch <= '9') {
        cout << "Digit";
    } else {
        cout << "Special Character";
    }
    return 0;
}
```

44. BMI Calculator Category:

C++

```
#include <iostream>
using namespace std;

int main() {
    float bmi = 22.5;
    if (bmi < 18.5) {
        cout << "Underweight";
    } else if (bmi <= 24.9) {
        cout << "Normal Weight";
    } else if (bmi <= 29.9) {
        cout << "Overweight";
    } else {
        cout << "Obese";
    }
    return 0;
}
```

45. Tax Bracket Calculator:

C++

```
#include <iostream>
using namespace std;

int main() {
    double income = 650000;
    if (income <= 250000) {
        cout << "No Tax";
    } else if (income <= 500000) {
        cout << "Tax Rate 5%";
    } else if (income <= 1000000) {
        cout << "Tax Rate 20%";
    } else {
        cout << "Tax Rate 30%";
    }
    return 0;
}
```

SECTION 4: Nested `if` Programs

A. Easy Nested `if` Programs

1. Check Positive Number and Even:

C++

```
#include <iostream>

using namespace std;

int main() {

    int num = 4;

    if (num > 0) {

        if (num % 2 == 0) {

            cout << "Positive Even Number";

        }

    }

    return 0;

}
```

2. Check Age and Voting Eligibility:

C++

```
#include <iostream>

using namespace std;

int main() {

    int age = 20;
```

```
    bool citizen = true;
    if (age >= 18) {
        if (citizen == true) {
            cout << "Eligible to vote";
        }
    }
    return 0;
}
```

3. Check Number Greater than 10 and 50:

C++

```
#include <iostream>
using namespace std;
int main() {
    int x = 60;
    if (x > 10) {
        if (x > 50) {
            cout << "Greater than 50";
        }
```

```
    }

    return 0;

}
```

4. Check Pass Marks with Distinction:

C++

```
#include <iostream>

using namespace std;

int main() {

    int marks = 90;

    if (marks >= 50) {

        if (marks >= 85) {

            cout << "Distinction";

        }

    }

    return 0;

}
```

5. Check Non-Zero and Positive:

C++

```
#include <iostream>

using namespace std;

int main() {

    int val = 12;

    if (val != 0) {

        if (val > 0) {

            cout << "Strictly Positive";

        }

    }

    return 0;

}
```

B. Normal Nested `if` Programs

6. Maximum of Three Numbers:

C++

```
#include <iostream>

using namespace std;

int main() {
```

```
    int a = 10, b = 25, c = 15;
    if (a >= b) {
        if (a >= c) {
            cout << "A is largest";
        } else {
            cout << "C is largest";
        }
    } else {
        if (b >= c) {
            cout << "B is largest";
        } else {
            cout << "C is largest";
        }
    }
    return 0;
}
```

7. Check Leap Year (Nested):

C++

```
#include <iostream>

using namespace std;

int main() {

    int year = 2024;

    if (year % 4 == 0) {

        if (year % 100 == 0) {

            if (year % 400 == 0) {

                cout << "Leap Year";

            } else {

                cout << "Not Leap Year";

            }

        } else {

            cout << "Leap Year";

        }

    } else {

        cout << "Not Leap Year";

    }

    return 0;
```

```
}
```

8. Check Login Credentials:

C++

```
#include <iostream>

using namespace std;

int main() {

    int user_id = 101, pass = 1234;

    if (user_id == 101) {

        if (pass == 1234) {

            cout << "Login Successful";

        } else {

            cout << "Incorrect Password";

        }

    } else {

        cout << "User Not Found";

    }

    return 0;

}
```

9. Check Divisibility by 2 and 3:

C++

```
#include <iostream>

using namespace std;

int main() {

    int n = 18;

    if (n % 2 == 0) {

        if (n % 3 == 0) {

            cout << "Divisible by both 2 and 3";

        } else {

            cout << "Divisible by 2 only";

        }

    } else {

        cout << "Not divisible by 2";

    }

    return 0;

}
```

10. Quadrant Detection:

C++

```
#include <iostream>

using namespace std;

int main() {

    int x = 5, y = -3;

    if (x > 0) {

        if (y > 0) {

            cout << "First Quadrant";

        } else {

            cout << "Fourth Quadrant";

        }

    } else {

        if (y > 0) {

            cout << "Second Quadrant";

        } else {

            cout << "Third Quadrant";

        }
```

```
    }

    return 0;

}
```

C. Advanced Nested `if` Program

11. Bank ATM Withdrawal System:

C++

```
#include <iostream>

using namespace std;

int main() {

    double balance = 5000.0;

    double amount = 2000.0;

    bool card_active = true;

    if (card_active == true) {

        if (amount <= balance) {

            balance -= amount;

            cout << "Transaction Successful";

        } else {

            cout << "Insufficient Balance";
```

```
        }
    } else {
        cout << "Card Blocked";
    }
    return 0;
}
```

12. Scholarship Eligibility Criteria:

C++

```
#include <iostream>
using namespace std;
int main() {
    float gpa = 3.8;
    int attendance = 90;
    bool discipline_clear = true;
    if (gpa >= 3.5) {
        if (attendance >= 85) {
            if (discipline_clear == true) {
                cout << "Scholarship Awarded";
```

```
            } else {
                cout << "Discipline Issue";
            }
        } else {
            cout << "Low Attendance";
        }
    } else {
        cout << "Low GPA";
    }
    return 0;
}
```

13. Income Tax Slabs with Surcharge:

C++

```
#include <iostream>
using namespace std;
int main() {
    double income = 1200000;
    if (income > 500000) {
```

```
        if (income > 1000000) {
            cout << "High Tax with Surcharge";
        } else {
            cout << "Standard Tax Slab 2";
        }
    } else {
        if (income > 250000) {
            cout << "Standard Tax Slab 1";
        } else {
            cout << "No Tax";
        }
    }
    return 0;
}
```

14. Triangle Type Validation:

C++

```
#include <iostream>
using namespace std;
```

```
int main() {
    int a = 60, b = 60, c = 60;
    if (a + b + c == 180) {
        if (a == b && b == c) {
            cout << "Equilateral Triangle";
        } else {
            if (a == b || b == c || a == c) {
                cout << "Isosceles Triangle";
            } else {
                cout << "Scalene Triangle";
            }
        }
    } else {
        cout << "Invalid Triangle";
    }
    return 0;
}
```

15. Advanced Admission Selection:

C++

```
#include <iostream>

using namespace std;

int main() {

    int test_score = 85;

    int interview_score = 90;

    int age = 21;

    if (age <= 25) {

        if (test_score >= 80) {

            if (interview_score >= 85) {

                cout << "Fully Selected";

            } else {

                cout << "Waiting List (Interview)";

            }

        } else {

            cout << "Rejected on Test Score";

        }

    } else {
```

```
        cout << "Overage";
    }
    return 0;
}
```

SECTION 5: `switch-case` Programs

A. Easy `switch-case` Programs

1. Day of the Week:

C++

```
#include <iostream>
using namespace std;
int main() {
    int day = 3;
    switch (day) {
        case 1: cout << "Monday"; break;
        case 2: cout << "Tuesday"; break;
        case 3: cout << "Wednesday"; break;
        case 4: cout << "Thursday"; break;
        case 5: cout << "Friday"; break;
```

```
        default: cout << "Weekend"; break;
    }
    return 0;
}
```

2. Vowel or Consonant Checker:

C++

```
#include <iostream>
using namespace std;
int main() {
    char ch = 'e';
    switch (ch) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
            cout << "Vowel";
            break;
```

```
        default:
            cout << "Consonant";
            break;
    }
    return 0;
}
```

3. Simple Traffic Light System:

C++

```
#include <iostream>
using namespace std;
int main() {
    char signal = 'R';
    switch (signal) {
        case 'R': cout << "Stop"; break;
        case 'Y': cout << "Get Ready"; break;
        case 'G': cout << "Go"; break;
        default: cout << "Invalid Signal"; break;
    }
```

```
    return 0;
}
```

4. Number Name (1 to 3):

C++

```
#include <iostream>
using namespace std;
int main() {
    int n = 2;
    switch (n) {
        case 1: cout << "One"; break;
        case 2: cout << "Two"; break;
        case 3: cout << "Three"; break;
        default: cout << "Out of Range"; break;
    }
    return 0;
}
```

5. Simple Choice Menu:

C++

```
#include <iostream>

using namespace std;

int main() {

    int choice = 1;

    switch (choice) {

        case 1: cout << "Play Game"; break;

        case 2: cout << "Settings"; break;

        case 3: cout << "Exit"; break;

        default: cout << "Invalid Choice"; break;

    }

    return 0;

}
```

B. Normal `switch-case` Programs

6. Basic Calculator:

C++

```
#include <iostream>

using namespace std;
```

```
int main() {
    char op = '+';
    double a = 10.0, b = 5.0;
    switch (op) {
        case '+': cout << a + b; break;
        case '-': cout << a - b; break;
        case '*': cout << a * b; break;
        case '/': cout << a / b; break;
        default: cout << "Invalid Operator"; break;
    }
    return 0;
}
```

7. Month Name from Number:

C++

```
#include <iostream>
using namespace std;
int main() {
    int m = 4;
```

```
    switch (m) {
        case 1: cout << "January"; break;
        case 2: cout << "February"; break;
        case 3: cout << "March"; break;
        case 4: cout << "April"; break;
        case 5: cout << "May"; break;
        case 6: cout << "June"; break;
        default: cout << "Other Month"; break;
    }
    return 0;
}
```

8. Grade Evaluation:

C++

```
#include <iostream>
using namespace std;
int main() {
    char grade = 'B';
    switch (grade) {
```

```
        case 'A': cout << "Excellent"; break;
        case 'B': cout << "Good"; break;
        case 'C': cout << "Average"; break;
        case 'D': cout << "Below Average"; break;
        case 'F': cout << "Fail"; break;
        default: cout << "Invalid Grade"; break;
    }
    return 0;
}
```

9. Language Selection Menu:

C++

```
#include <iostream>
using namespace std;
int main() {
    int lang = 2;
    switch (lang) {
        case 1: cout << "English Selected"; break;
        case 2: cout << "Urdu Selected"; break;
```

```
        case 3: cout << "Arabic Selected"; break;
        default: cout << "Default English"; break;
    }
    return 0;
}
```

10. Direction Finder:

C++

```
#include <iostream>
using namespace std;
int main() {
    char dir = 'N';
    switch (dir) {
        case 'N': cout << "North"; break;
        case 'S': cout << "South"; break;
        case 'E': cout << "East"; break;
        case 'W': cout << "West"; break;
        default: cout << "Invalid Direction"; break;
    }
```

```
    return 0;
}
```

C. Advanced `switch-case` Programs

11. Days in a Month Calculator:

C++

```
#include <iostream>
using namespace std;
int main() {
    int month = 2, year = 2024;
    switch (month) {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            cout << "31 Days";
```

```
        break;
    case 4:
    case 6:
    case 9:
    case 11:
        cout << "30 Days";
        break;
    case 2:
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            cout << "29 Days";
        } else {
            cout << "28 Days";
        }
        break;
    default:
        cout << "Invalid Month";
        break;
}
```

```
    return 0;
}
```

12. Coffee Vending Machine Menu:

C++

```
#include <iostream>
using namespace std;
int main() {
    int choice = 2;
    int size = 1;
    switch (choice) {
        case 1:
            switch (size) {
                case 1: cout << "Small Espresso"; break;
                case 2: cout << "Large Espresso"; break;
            }
            break;
        case 2:
            switch (size) {
```

```
            case 1: cout << "Small Cappuccino"; break;
            case 2: cout << "Large Cappuccino"; break;
        }
        break;
    default:
        cout << "Invalid Selection";
        break;
  }
  return 0;
}
```

13. Geometric Shape Area Selector:

C++

```
#include <iostream>
using namespace std;
int main() {
  int shape_code = 1;
  double dim1 = 5.0, dim2 = 10.0;
  switch (shape_code) {
```

```
        case 1: cout << "Circle Area: " << 3.1416 * dim1 * dim1; break;
        case 2: cout << "Rectangle Area: " << dim1 * dim2; break;
        case 3: cout << "Triangle Area: " << 0.5 * dim1 * dim2; break;
        default: cout << "Invalid Shape Code"; break;
    }
    return 0;
}
```

14. User Role Permission System:

C++

```
#include <iostream>
using namespace std;
int main() {
    int role = 2;
    switch (role) {
        case 1: cout << "Full Administrator Access"; break;
        case 2: cout << "Manager: Read and Write Access"; break;
        case 3: cout << "Editor: Modify Content Only"; break;
        case 4: cout << "Guest: Read Only Access"; break;
```

```
        default: cout << "Access Denied"; break;
    }
    return 0;
}
```

15. Advanced Currency Converter:

C++

```
#include <iostream>
using namespace std;
int main() {
    int currency_code = 2;
    double amount_pkr = 28000.0;
    switch (currency_code) {
        case 1: cout << "USD: " << amount_pkr / 280.0; break;
        case 2: cout << "EUR: " << amount_pkr / 300.0; break;
        case 3: cout << "GBP: " << amount_pkr / 350.0; break;
        case 4: cout << "SAR: " << amount_pkr / 75.0; break;
        default: cout << "Invalid Currency Code"; break;
    }
```

```
    return 0;

}
```